ASSIGNMENT 1 – CO1 | TASK 4

Control Implicit Conversions

Scenario: Namma Metro Smart-Card Fare System

Objective / explanation

The program demonstrates integer division, explicit conversion to double, char promotion, conversion to bool, and signed/unsigned comparison.

C++ source code

```
#include &lt;iostream&gt;
using namespace std;

int main() {
    int a = 7, b = 2;
    double c = a / b;
    double d = static_cast&lt;double&gt;(a) / b;
    cout &lt;&lt; c &lt;&lt; ' ' &lt;&lt; d &lt;&lt; '\n';
    char ch = 'A' + 2;
    cout &lt;&lt; ch &lt;&lt; ' ' &lt;&lt; ch + 1 &lt;&lt; '\n';
    bool flag = 5;
    cout &lt;&lt; flag &lt;&lt; ' ';
    cout &lt;&lt; boolalpha &lt;&lt; flag &lt;&lt; '\n';
    unsigned int u = 1;
    int s = -1;
    cout &lt;&lt; (s &lt; u ? "s &lt; u" : "s &gt;= u") &lt;&lt; '\n';
}
```

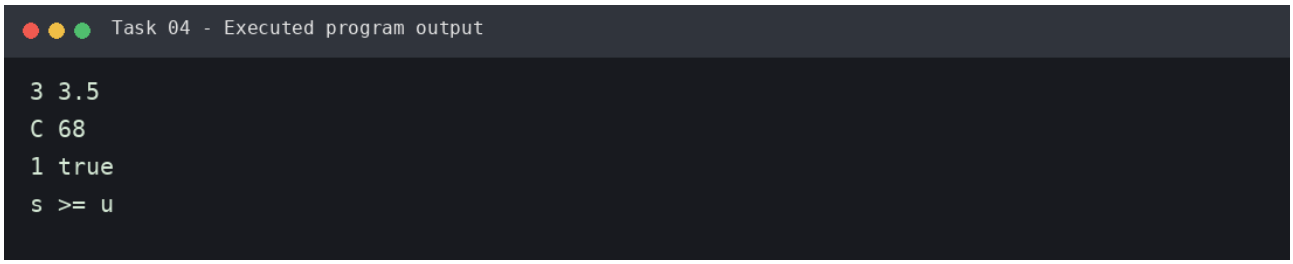
Compilation

```
g++ -std=c++17 -Wall -Wextra -pedantic main.cpp -o program
```

Actual captured program output

```
3 3.5
C 68
1 true
s &gt;= u
```

Output screenshot



Analysis / complexity

Integer division produces 3 before conversion to double. static_cast<double>(a) gives 3.5. char is promoted to int in ch+1. 5 converts to true. In the mixed signed/unsigned comparison, -1 converts to a large unsigned value.

Evidence note: Output shown is captured by executing this source in the generation environment. Recompile and run on your own system, and use your own screenshots if your instructor requires evidence from your computer.